

# Audrey Shih

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## EDUCATION

|                                                               |                      |
|---------------------------------------------------------------|----------------------|
| <b>Stanford University</b>                                    | <b>Stanford, CA</b>  |
| Ph.D., Chemical Engineering, GPA 3.96                         | Expected June 2025   |
| M.S., Chemical Engineering                                    | 2020 – 2023          |
| <b>Princeton University</b>                                   | <b>Princeton, NJ</b> |
| B.S.E., Chemical and Biological Engineering, <i>cum laude</i> | 2016 – 2020          |
| Minor, Materials Science and Engineering                      |                      |

## EXPERIENCE

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| <b>Stanford University</b> , Stanford, CA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2021 – Present |
| <i>Doctoral Researcher</i>   PI: Gerald G. Fuller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |
| <ul style="list-style-type: none"><li>Developed and optimized a portable magnetic stress rheometer using CAD-designed components and Raspberry Pi, reducing device costs from ~\$90k to \$160 for use in medical settings. Applied hands-on engineering, testing, and troubleshooting to enhance real-time fluid flow predictions for abscess drainage and biopolymeric materials.</li><li>Engineered 3D bioprinting systems for rheological analysis of bioinks in diffusion-driven crosslinking, solving technical challenges related to optimizing mechanical properties of printed complex biological tissues.</li><li>Managed lab safety protocols and compliance as designated lab safety officer, attending quarterly environmental health and safety meetings with University officials to ensure regulatory adherence in research environments.</li><li>Instructed and mentored both undergrad and grad students in <i>Mechanics of Soft Matter: Rheology (CHEMENG 470)</i>.</li></ul> |                |
| <b>Stanford University</b> , Stanford, CA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2021           |
| <i>Doctoral Researcher</i>   PI: Joseph M. DeSimone                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |
| <ul style="list-style-type: none"><li>Developed and optimized high-resolution 3D Continuous Liquid Interface Production (CLIP) printing through experiments, SEM imaging, and simulations of optics and photopolymerization transport and kinetics.</li><li>Utilized CAD and 3D-slicing software to test and optimize 3D printer settings, enhancing resolution and precision in additive manufacturing processes.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |
| <b>Princeton University</b> , Princeton, NJ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2018 – 2020    |
| <i>Undergraduate Researcher</i>   PI: Sujit S. Datta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |
| <ul style="list-style-type: none"><li>Investigated elastic turbulence in polymer flow by CAD designing 3D-printed microfluidic model porous channels.</li><li>Featured in ACEE's first ever <a href="#">spotlight article</a> on undergraduate contributions to environmental studies.</li><li>Won Outstanding Thesis Award, thesis featured in <a href="#">profile article</a> by the School of Engineering and Applied Science.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |
| <b>TU-Dortmund, Fakultät Bio und Chemieingenieurwesen</b> , Dortmund, Germany                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2018           |
| <i>Research Intern</i>   Armin Quentmeier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |
| <ul style="list-style-type: none"><li>Commandeered the Biochemical and Chemical Engineering department's contribution to the 2018 Engineering Meets Art exhibition, designed project involving photographs of microorganisms and colonies.</li><li>Wrote 40-page manual of protocols for inoculation and dilution techniques, instructions for using cameras, lenses, microscopes, and Adobe Photoshop Elements, and analyzed microscopic images using various software.</li><li>Final projects displayed at exhibition from May – June 2019.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |
| <b>Princeton University</b> , Princeton, NJ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2017 – 2018    |
| <i>Undergraduate Researcher</i>   PI: Celeste M. Nelson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |
| <ul style="list-style-type: none"><li>Analyzed role of tissue mechanics in breast cancer tumor development using immunostaining techniques.</li><li>Synthesized and crosslinked polyacrylamide gels, passaged and stained cells, analyzed results with mass spectrometry, Western blot, and confocal microscopy.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |

## SKILLS

*Technical/Laboratory:* Fluid mechanics; rheology; polymers; CAD (Fusion 360, Onshape); 3D printing (SLA, FDM, CLIP); hardware/embedded systems (PCB assembly, Raspberry Pi); SEM; GPC; DMA; PCR; cell culture; immunostaining; confocal microscopy imaging

*Software:* Java, MATLAB, Python; ImageJ; TRIOS; PIV; Adobe Illustrator, Photoshop; LaTeX; Microsoft Office

## PUBLICATIONS & PATENTS

- **Shih, A.**, Chung, S. J., Brunel, L. G., Christakopoulos, F., Heilshorn, S. C., Fuller, G. G., *In situ* rheological measurements of bioink crosslinking. *In preparation*.
- **Shih, A.**, Chung, S. J., Shende, O. B., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., [Viscoelastic measurements of abscess fluids using a magnetic stress rheometer](#). *Physics of Fluids* 36 (11): 111919 (2024), *Cover Article*.
- Cai, P. C., Braunreuther, M., **Shih, A.**, Spakowitz, A. J., Fuller, G. G., Heilshorn, S. C., [Air-liquid intestinal cell culture allows in situ rheological characterization of intestinal mucus](#). *APL Bioengineering* 8 (2): 026112 (2024).
- DeSimone, J. M., Jacobson, G. B., Dulay, M. T., Lee, B. J., Hsiao, K., Rajesh, N., Driskill, M. M., **Shih, A.**, et. al., *Polymeric microstructures and systems and methods for making same*. [Patent number WO2023049267A1 \(2023\)](#).
- Hsiao, K., Lee, B. J., Samuelsen, T., Lipkowitz, G., Kronenfeld, J. M., Ilyn, D., **Shih, A.**, et al., [Single-digit-micrometer-resolution continuous liquid interface production](#). *Science Advances* 8, eabq2846 (2022).
- Browne, C. A., **Shih, A.**, Datta, S. S., [Bistability in the Unstable Flow of Polymer Solutions in Porous Media](#). *Journal of Fluid Mechanics* 890:A2 (2020).
- Browne, C. A., **Shih, A.**, Datta, S. S., [Pore-Scale Flow Characterization of Polymer Solutions in Microfluidic Porous Media](#). *Small* 16, 1903944 (2019).

## SELECTED AWARDS

|            |                                                                                          |
|------------|------------------------------------------------------------------------------------------|
| 2024       | Excellence in Fluid Mechanics Research and Oral Presentation Award, AIChE Annual Meeting |
| 2024, 2020 | Sigma Xi Scientific Research Honor Society Nominee                                       |
| 2023       | Chemical Engineering Department Service Leadership Award                                 |
| 2021       | Judges' Vote and Audience Choice poster awards, Stanford MIPS Retreat, Woodside, CA      |
| 2020       | National Science Foundation (NSF) Graduate Research Fellowship                           |
| 2020       | Lore von Jaskowsky Memorial Prize, Princeton School of Engineering and Applied Sciences  |
| 2020       | Materials Science and Engineering Department Outstanding Senior Thesis Award             |
| 2019       | PSEG Best Poster Award, ACEE Annual Meeting, Princeton, NJ                               |

## CONFERENCE TALKS

- **Shih, A.**, Chung, S. J., Shende, O. B., Brunel, L. G., Christakopoulos, F., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. American Society of Chemical Engineers Annual Meeting (AIChE 2024), San Diego, CA, Oct. 2024.
- **Shih, A.**, Chung, S. J., Shende, O. B., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Rheological Characterization of Biological Fluids*. NETZSCH Introduction to Rheology and Thermal Analysis Workshop, Palo Alto, CA, Oct. 2024.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. American Society of Chemical Engineers Annual Meeting (AIChE 2023), Orlando, FL, Nov. 2023.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. XIXth International Congress on Rheology (ICR 2023), Athens, Greece, Aug. 2023.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. Society of Rheology Annual Meeting (SOR 2022), Chicago, IL, Oct. 2022.

## LEADERSHIP & MENTORSHIP

|                                           |                |
|-------------------------------------------|----------------|
| <b>Stanford University</b> , Stanford, CA | 2022 – Present |
| <i>Research Mentor</i>                    |                |

- Mentored two undergraduate students (Stanford University, Shenzhen University) and one high school intern (Germantown Friends School), teaching CAD, Python, MATLAB, ImageJ, fluid mechanics, and polymer theory.
- Provided hands-on guidance in hardware, coding, and data analysis, fostering technical skills and problem-solving abilities for real-world research projects in rheology, medicine, and 3D bioprinting.

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|-------------------------------------------------------------------------|----------------|
| <b>Graduate Life Office</b> , <b>Stanford University</b> , Stanford, CA | 2021 – Present |
| <i>Community Associate</i>                                              |                |

- Coordinated 12 events annually to foster community among graduate student population, managing a \$3,600 budget for diverse and inclusive programs, including welcome, orientation, and multicultural initiatives.
- Advised graduate residents on support services and professional development, utilizing annual training on suicide prevention, servant leadership, bias intervention, and sexual harassment prevention.

**Self-employed**

2018 – present

*STEM tutor and clarinet instructor*

- Taught math and chemistry to students from ages 10-17 and clarinet performance to adult beginner students.

**Graduate Student Action Committee, Stanford University, Stanford, CA**

2022 – 2023

*Social Events Chair* | Department of Chemical Engineering

- Served as liaison between ChemE student community and administration through quarterly meetings.
- Spearheaded multi-departmental events to expand the department's community, promoting social and collaborative connections across various department programs.

**Stanford University, Stanford, CA**

2022 – 2023

*First-Year Mentor* | Department of Chemical Engineering

- Guided first-year students in selecting labs, managing coursework, and preparing for qualification exams, sharing resources and support through three meetings per quarter.

**Stanford Splash, Stanford, CA**

2023

*Teacher Volunteer*

- Taught hands-on workshop on polymer crosslinking to middle- and high-school students in two classes.

**PROFESSIONAL AFFILIATIONS**

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Society of Rheology, American Institute of Chemical Engineers